

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Debugging & Optimization Task

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	Total Marks:	100
CO2 Statement:	Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.		
Student Name:	G.JEEVESH	Reg. No.:	192511440
Section / Batch:	B	Date:	22/07/2026

Code of Conduct

I G.JEEVESH(192511440)_____ (Name / Reg No)
certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: G.Jeevesh_____

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 – 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
GRAND TOTAL:			100 Marks

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well-documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well-justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____



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DS PROGRAMMING



Multiple Choice Questions

0/25 answered

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)
OPEN
✓ Completed

Debugging & Optimization Task

10/10 answered

Review →

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)
OPEN
✓ Completed

Debugging & Optimization Task

10/10 answered

Review →

CO1_AT1_C CODE OUTPUT PREDICTION
OPEN
✓ Completed

C CODE OUTPUT PREDICTION

10/10 answered

Review →

QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack
5 marksQ6
Linked List
5 marksQ7
Stack
5 marksQ8
Linked List Data Structures
5 marksQ9
Linked List
5 marksQ10
Queue
5 marks

10/10 answered

Q1 Binary Search

5 marks

Identify the error in the following snippet.

```
int binarySearch(int arr[], int n, int key) {  
    int low = 0, high = n - 1;  
    while(low < high) {  
        int mid = (low + high) / 2;  
        if(arr[mid] == key)  
            return mid;  
        else if(arr[mid] < key)  
            low = mid;  
        else  
            high = mid;  
    }  
    return -1;  
}
```

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int binarySearch(int arr[], int n, int key)  
2 {  
3     int low = 0, high = n - 1;  
4  
5     while (low <= high)  
6     {  
7         int mid = low + (high - low) / 2;  
8  
9         if (arr[mid] == key)  
10            return mid;  
11        else if (arr[mid] < key)  
12            low = mid + 1;  
13        else  
14            high = mid - 1;  
15    }  
16    return -1;  
17 }
```

Test Input

stdin input (optional)

Output

⌚ Pending manual review

QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack
5 marksQ6
Linked List
5 marksQ7
Stack
5 marksQ8
Linked List Data Structures
5 marksQ9
Linked List
5 marksQ10
Queue
5 marks

Q2 Stack

5 marks

Find the error in the stack deletion operation. Justify it.

```
int pop() {  
    if(top == 0) {  
        printf("Underflow");  
        return -1;  
    }  
    return stack[top--];  
}
```

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 int stack[100];  
4 int top = -1;  
5  
6 int pop() {  
7     if (top == -1) {  
8         printf("Underflow\n");  
9         return -1;  
10    }  
11    return stack[top--];  
12 }  
13  
14 int main() {  
15     stack[++top] = 10;  
16     stack[++top] = 20;  
17     stack[++top] = 30;  
18  
19     printf("%d\n", pop());  
20  
21     return 0;  
22 }
```

Test Input

Push: 10
Push: 20
Push: 30
Pop

Output

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

[← Back](#)

QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack
5 marks

Q6

Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List Data Structures
5 marks

Q9

1 feedback 1 like



Q3 Stack

5 marks

Check the snippet and identify the logical error.

```
int stack[5];
int top;
void init() {
    top = 0;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 int stack[5];
4 int top;
5
6 void init() {
7     top = -1;
8 }
9
10 int main() {
11     init();
12     printf("Top = %d\n", top);
13     return 0;
14 }
```

Test Input

No input

Output

Pending manual review

Submitted



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DS PROGRAMMING

JEEVESH G
192511440RB

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

[← Back](#)

QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack
5 marks

Q6

Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List Data Structures
5 marks

Q9

Linked List
5 marks

Q4 Stack

5 marks

Identify the errors in the following snippet.

```
if(ch == ')') {
    while(stack[top] != ')') {
        postfix[j++] = pop();
    }
    pop();
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 if(ch == ')')
2 {
3     while (top != -1 && stack[top] != '(')
4     {
5         postfix[j++] = pop();
6     }
7
8     if (top == -1)
9     {
10        printf("Invalid Expression");
11        return;
12    }
13
14    pop();
15 }
```

Test Input

(A+B)*C

Output

Pending manual review

Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack
5 marksQ6
Linked List
5 marksQ7
Stack
5 marksQ8
Linked List Data Structures
5 marksQ9
Linked List
5 marks

Q5 Stack

5 marks

In balancing Bracket, you identify the error.

```
if((ch == '[' && stack[top] == '[') ||  
(ch == ']' && stack[top] == '[')) {  
    pop();  
} else {  
    return 0;  
}
```

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 if (top != -1 &&  
2 ((ch == '[' && stack[top] == '[') ||  
3 (ch == ']' && stack[top] == '[') ||  
4 (ch == '[' && stack[top] == '[')))  
5 {  
6     pop();  
7 }  
8 else  
9 {  
10     return 0;  
11 }
```

Test Input

{[()]}

Output

⌚ Pending manual review

✔ Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack
5 marksQ6
Linked List
5 marksQ7
Stack
5 marksQ8
Linked List Data Structures
5 marksQ9
Linked List
5 marks

Q6 Linked List

5 marks

What is the issue?

```
temp = head  
while temp.next != NULL:  
    print(temp.data)  
temp = temp.next
```

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 struct Node {  
4     int data;  
5     struct Node *next;  
6 };  
7  
8 void display(struct Node *head)  
9 {  
10     struct Node *temp = head;  
11  
12     while (temp != NULL)  
13     {  
14         printf("%d ", temp->data);  
15         temp = temp->next;  
16     }  
17 }
```

Test Input

Linked List: 10 -> 20 -> 30 -> NULL

Output

⌚ Pending manual review

✔ Submitted

QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack
5 marks

Q6

Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List Data Structures
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

Q7 Stack

5 marks

Point out the error.
POP(stack):
if top == 0:
print("Underflow")
else:
return stack[top--]

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 int stack[100];
4 int top = -1;
5
6 int pop()
7 {
8     if (top == -1)
9     {
10         printf("Underflow\n");
11         return -1;
12     }
13     return stack[top--];
14 }
15
16 int main()
17 {
18     stack[++top] = 10;
19     stack[++top] = 20;
20
21     printf("%d\n", pop());
22
23     return 0;
24 }
```

Test Input

Push 10
Push 20
Pop

Output



⌚ Pending manual review



Search



QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack
5 marks

Q6

Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List Data Structures
5 marks

Q9

Linked List
5 marks

Q10



Q8 Linked List Data Structures

5 marks

Identify the errors in the following code

```
while current != NULL:
    current.next = prev
    prev = current
    current = current.next
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 struct Node *next;
2
3 while (current != NULL)
4 {
5     next = current->next;
6     current->next = prev;
7     prev = current;
8     current = next;
9 }
10 head = prev;
```

Test Input

Linked List:
10 -> 20 -> 30 -> 40 -> NULL

Output



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📝 Submitted

QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack
5 marks

Q6

Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List Data Structures
5 marks

Q9 Linked List

5 marks

What is the error in the below code?

```
new.next = head
head = new
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 struct Node *new = (struct Node *)malloc(sizeof(struct Node));
2
3 if (new == NULL)
4 {
5     printf("Memory allocation failed");
6     return;
7 }
8
9 new->data = value;
10 new->next = head;
11 head = new;
```

Test Input

Enter value: 25

Output



🕒 Pending manual review

📝 Submitted

QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack
5 marks

Q6

Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List Data Structures
5 marks

Q9

Linked List
5 marks

Q10 Queue

5 marks

What is missing in the below pseudocode?

```
DEQUEUE(Q):  
if front == -1:  
    print("Underflow")  
else:  
    return Q[front++]
```

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 DEQUEUE(Q)  
2 {  
3     if (front == -1)  
4     {  
5         printf("Underflow");  
6         return -1;  
7     }  
8  
9     int item = Q[front++];  
10  
11     if (front > rear)  
12     {  
13         front = rear = -1;  
14     }  
15  
16     return item;  
17 }
```

Test Input

Queue: 10 20 30
Dequeue

Output



⌚ Pending manual review

📌 Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

[← Back](#)

QUESTIONS

Q1

Stack Notations - Postfix
Expression
5 marks

Q2

Circular Queue
5 marks

Q3

Binary Search
5 marks

Q4

Queue
5 marks

Q5

Queue
5 marks

Q6

Doubly Linked List
5 marks

Q7

Stack
5 marks

Q8

Stack
5 marks

Q9

Circular Queue
5 marks

Q10

Stack



Q1 Stack Notations - Postfix Expression

5 marks

Find the bug in the evaluation of a Postfix expression.

```
while (expression[i] != '\0') {  
    if (isdigit(expression[i])) push(expression[i] - '0');  
    else {  
        int op1 = pop();  
        int op2 = pop();  
        // perform operation: result = op1 + op2;  
        push(result);  
    }  
}
```

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 #include <ctype.h>  
3  
4 #define MAX 100  
5  
6 int stack[MAX];  
7 int top = -1;  
8  
9 void push(int x)  
10 {  
11     stack[++top] = x;  
12 }  
13  
14 int pop()  
15 {  
16     return stack[top--];  
17 }  
18
```

Test Input

23*54*+9-

Output



⌚ Pending manual review

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

[← Back](#)

QUESTIONS

Q1
Stack Notations - Postfix
Expression
5 marks**Q2**
Circular Queue
5 marks**Q3**
Binary Search
5 marks**Q4**
Queue
5 marks**Q5**
Queue
5 marks**Q6**
Doubly Linked List
5 marks**Q7**
Stack
5 marks**Q8**
Stack
5 marks**Q9**
Circular Queue
5 marks**Q10**
Stack**Q2 Circular Queue**

5 marks

Identify the bug in this Circular Queue dequeue logic.

```
int dequeue() {  
    if (front == -1) return -1;  
    int data = queue[front];  
    front = (front + 1) % MAX;  
    return data;  
}
```

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 #define MAX 5  
3  
4 int queue[MAX];  
5 int front = -1, rear = -1;  
6  
7 void enqueue(int data)  
8 {  
9     if ((rear + 1) % MAX == front)  
10    {  
11        printf("Queue is full\n");  
12        return;  
13    }  
14  
15    if (front == -1)  
16        front = 0;  
17  
18    rear = (rear + 1) % MAX;  
19    queue[rear] = data;  
20 }  
21  
22 int dequeue()
```

Test Input

stdin input (optional)

Output

 Pending manual review

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

[← Back](#)

QUESTIONS

Q1
Stack Notations - Postfix
Expression
5 marks**Q2**
Circular Queue
5 marks**Q3**
Binary Search
5 marks**Q4**
Queue
5 marks**Q5**
Queue
5 marks**Q6**
Doubly Linked List
5 marks**Q7**
Stack
5 marks**Q8**
Stack
5 marks**Q9**
Circular Queue
5 marks**Q10**
Stack**Q3 Binary Search**

5 marks

Identify the bug in this Recursive Binary Search.

```
int binarySearch(int arr[], int l, int r, int x) {  
    int mid = l + (r - l) / 2;  
    if (arr[mid] == x) return mid;  
    if (arr[mid] > x) return binarySearch(arr, l, mid, x);  
    return binarySearch(arr, mid, r, x);  
}
```

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 int binarySearch(int arr[], int l, int r, int x)  
4 {  
5     if (l > r)  
6         return -1;  
7  
8     int mid = l + (r - l) / 2;  
9  
10    if (arr[mid] == x)  
11        return mid;  
12  
13    if (arr[mid] > x)  
14        return binarySearch(arr, l, mid - 1, x);  
15  
16    return binarySearch(arr, mid + 1, r, x);  
17 }  
18  
19 int main()  
20 {  
21     int arr[] = {10, 20, 30, 40, 50};  
22     int n = 5, x = 40;
```

Test Input

stdin input (optional)

Output

Element found at index 3

Pending manual review

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

[← Back](#)

QUESTIONS

Q1
Stack Notations - Postfix
Expression
5 marks**Q2**
Circular Queue
5 marks**Q3**
Binary Search
5 marks**Q4**
Queue
5 marks**Q5**
Queue
5 marks**Q6**
Doubly Linked List
5 marks**Q7**
Stack
5 marks**Q8**
Stack
5 marks**Q9**
Circular Queue
5 marks**Q4 Queue**

5 marks

Find the error in this Deque insertFront operation.

```
void insertFront(int data) {  
    if (front == 0) front = MAX - 1;  
    else front = front - 1;  
    deque[front] = data;  
}
```

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 void insertFront(int data) {  
2     if (front == -1) {  
3         front = rear = 0;  
4     }  
5     else if (front == 0) {  
6         front = MAX - 1;  
7     }  
8     else {  
9         front = front - 1;  
10    }  
11    deque[front] = data;  
12 }
```

Test Input

stdin input (optional)

Output

```
Compilation Error:  
C:\Windows\TEMP\sandbox_19251144  
ORB_6a794ae60f1753_41121518\main  
.c: In function 'insertFront':  
C:\Windows\TEMP\sandbox_19251144  
ORB_6a794ae60f1753_41121518\main.c:  
11: error: 'deque' is undeclared identifier
```

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Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1
Stack Notations - Postfix
Expression
5 marks**Q2**
Circular Queue
5 marks**Q3**
Binary Search
5 marks**Q4**
Queue
5 marks**Q5**
Queue
5 marks**Q6**
Doubly Linked List
5 marks**Q7**
Stack
5 marks**Q8**
Stack
5 marks**Q9**
Circular Queue
5 marks**Q10**
Stack**Q5 Queue**

5 marks

Identify the error in the Queue enqueue logic.

```
void enqueue(int data) {  
    if (rear == MAX) return;  
    queue[rear++] = data;  
}
```

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 #define MAX 5  
4  
5 int queue[MAX];  
6 int front = 0;  
7 int rear = -1;  
8  
9 void enqueue(int data)  
10 {  
11     if (rear == MAX - 1)  
12     {  
13         printf("Queue Overflow\n");  
14         return;  
15     }  
16  
17     queue[++rear] = data;  
18 }  
19  
20 void dequeue()  
21 {  
22     if (front > rear)  
23     {  
24         printf("Queue Underflow\n");
```

Test Input

stdin input (optional)

Output

```
Queue elements: 10 20 30 40 50  
Deleted element: 10  
Deleted element: 20  
Queue elements: 30 40 50  
Queue Overflow  
Queue elements: 30 40 50
```

🚫 Pending manual review

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

[← Back](#)

QUESTIONS

Q1
Stack Notations - Postfix
Expression
5 marks**Q2**
Circular Queue
5 marks**Q3**
Binary Search
5 marks**Q4**
Queue
5 marks**Q5**
Queue
5 marks**Q6**
Doubly Linked List
5 marks**Q7**
Stack
5 marks**Q8**
Stack
5 marks**Q9**
Circular Queue
5 marks**Q10**
Stack**Q6 Doubly Linked List**

5 marks

Identify the error in this Doubly Linked List node deletion.

```
void deleteNode(Node* del) {  
    if (del->prev != NULL)  
        del->prev->next = del->next;  
    if (del->next != NULL)  
        del->next->prev = del->prev;  
}
```

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 #include <stdlib.h>  
3  
4 struct Node {  
5     int data;  
6     struct Node *prev;  
7     struct Node *next;  
8 };  
9  
10 struct Node *head = NULL;  
11  
12 // Insert node at end  
13 void insert(int data) {  
14     struct Node *newNode = (struct Node *)malloc(sizeof(struct Node));  
15     newNode->data = data;  
16     newNode->prev = NULL;  
17     newNode->next = NULL;  
18  
19     if (head == NULL) {  
20         head = newNode;  
21         return;  
22     }
```

Test Input

Enter number of nodes: 5
Enter elements:
10 20 30 40 50
Enter element to delete: 30

Output

 Pending manual review

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

[← Back](#)

QUESTIONS

Q1
Stack Notations - Postfix Expression
5 marks**Q2**
Circular Queue
5 marks**Q3**
Binary Search
5 marks**Q4**
Queue
5 marks**Q5**
Queue
5 marks**Q6**
Doubly Linked List
5 marks**Q7**
Stack
5 marks**Q8**
Stack
5 marks**Q9**
Circular Queue
5 marks**Q10**
Stack**Q7 Stack**

5 marks

Identify the issue in the below pseudocode.
POP():
if top == 0 → Underflow

🌟 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #define MAX 5
3
4 int stack[MAX];
5 int top = -1;
6
7 // Push operation
8 void push(int value)
9 {
10     if (top == MAX - 1)
11     {
12         printf("Stack Overflow\n");
13         return;
14     }
15     stack[++top] = value;
16 }
17
18 // Pop operation
19 void pop()
20 {
21     if (top == -1)
22     {
23         printf("Stack Underflow\n");
24         return;
25     }
26 }
```

Test Input

Enter number of elements: 4
Enter elements:
10 20 30 40

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1
Stack Notations - Postfix
Expression
5 marks**Q2**
Circular Queue
5 marks**Q3**
Binary Search
5 marks**Q4**
Queue
5 marks**Q5**
Queue
5 marks**Q6**
Doubly Linked List
5 marks**Q7**
Stack
5 marks**Q8**
Stack
5 marks**Q9**
Circular Queue
5 marks**Q10****Q8 Stack**

5 marks

Point out the error.
if top == MAX:
Overflow

🔍 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #define MAX 5
3
4 int stack[MAX];
5 int top = -1;
6
7 void push(int value)
8 {
9     if (top == MAX - 1)
10    {
11        printf("Stack Overflow\n");
12        return;
13    }
14
15    stack[++top] = value;
16    printf("%d pushed\n", value);
17 }
18
19 void display()
20 {
21     if (top == -1)
22     {
23         printf("Stack is empty\n");
24         return;
25     }
26 }
```

Test Input

Enter number of elements: 6
Enter elements:
10 20 30 40 50 60

Output

Enter number of elements: Enter
elements:
304 pushed
304 pushed
304 pushed
304 pushed

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1
Stack Notations - Postfix
Expression
5 marks**Q2**
Circular Queue
5 marks**Q3**
Binary Search
5 marks**Q4**
Queue
5 marks**Q5**
Queue
5 marks**Q6**
Doubly Linked List
5 marks**Q7**
Stack
5 marks**Q8**
Stack
5 marks**Q9**
Circular Queue
5 marks**Q10****Q9 Circular Queue**

5 marks

Identify the issue in the following snippet for Circular Queue in Traffic Signal.

```
rear = (rear + 1) % MAX;
queue[rear] = vehicle;
if(rear == front){
    printf("Queue Full");
}
```

🔍 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #define MAX 5
3
4 int queue[MAX];
5 int front = -1, rear = -1;
6
7 void enqueue(int vehicle)
8 {
9     // Check queue full before insertion
10    if ((rear + 1) % MAX == front)
11    {
12        printf("Queue Full\n");
13        return;
14    }
15
16    // First element
17    if (front == -1)
18        front = 0;
19
20    rear = (rear + 1) % MAX;
21    queue[rear] = vehicle;
22 }
```

Test Input

Enter number of vehicles: 6
Enter vehicle numbers:
101 102 103 104 105 106

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

[← Back](#)

QUESTIONS

- Q1**
Stack Notations - Postfix Expression
5 marks
- Q2**
Circular Queue
5 marks
- Q3**
Binary Search
5 marks
- Q4**
Queue
5 marks
- Q5**
Queue
5 marks
- Q6**
Doubly Linked List
5 marks
- Q7**
Stack
5 marks
- Q8**
Stack
5 marks
- Q9**
Circular Queue
5 marks
- Q10**

Q10 Stack

5 marks

Point out the error?

```
PEEK(stack);  
return stack[top+1]
```

✳ **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 #define MAX 5  
3  
4 int stack[MAX];  
5 int top = -1;  
6  
7 void push(int value)  
8 {  
9     if (top == MAX - 1)  
10    {  
11        printf("Stack Overflow\n");  
12        return;  
13    }  
14  
15    stack[++top] = value;  
16 }  
17  
18 void peek()  
19 {  
20     if (top == -1)  
21     {  
22         printf("Stack is empty\n");  
23         return;  
24     }  
25  
26     printf("Top element: %d\n", stack[top]);
```

Test Input

Enter number of elements: 4
Enter elements:
10 20 30 40

Output



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